

TASK DESCRIPTION DOCUMENT

[Instructions: There is no page limit to this template. Instructional text may be deleted. Provide a detailed task breakdown by phase (if applicable), citing specific tasks and their connection to the interim milestones and metrics, as applicable. Do not include proprietary information. For each defined task/subtask, provide:

- A general description of the objective.
- A detailed description of the approach to be taken to accomplish each defined task/subtask.
- Identification of any tasks/subtasks that will involve human subjects or animals.
- Identification of any tasks/subtasks that will be performed on campus at a university.
- Identification (by name) of the primary organization (prime contractor, subawardee(s), consultant(s)) responsible for task/subtask execution.
- Measurable milestones (e.g., a deliverable, demonstration, or other event/activity that marks task progress or completion).
- A definition of all deliverables (e.g., data, reports, software) to be provided to the Government in support of the proposed tasks/subtasks.]

The objective of this proposal is: Develop and empirically validate a composed-operator governance framework $T = L_{14} \circ \dots \circ L_1$ for multi-agent AI communication dynamics in scientific discovery tasks. The framework maps agent communication acts to points in a Poincaré ball, measures safety by hyperbolic distance from the safe-operation origin, and enforces five physical axioms across 14 layers. Phase I demonstrates the framework on NMR spectroscopy task families (Karplus equation rediscovery) using ChemBERTa-77M and Qwen2.5-Coder-0.5B-Instruct as small science models.

Objective	Establish the Phase I mathematical framework, science subdomains, foundation models, baseline protocol, personnel assignments, initial evaluation milestones, and IV&V artifact structure	2361 E 5th Ave, Port Angeles, WA 98362 (remote / local)
Work Description	Lead: SCBE / Issac Davis (Prime) Support: Hoags Inc. (limited to Sub Background IP review if needed)	
Primary Organization Responsible	SCBE / Issac Davis (Prime)	
Human Subjects or Animal Research?	No	
Associated Milestones	Kickoff briefing; foundation model and latent-space access plan; baseline protocol description; initial TA1 rubric mapping; personnel level-of-effort confirmation; draft IV&V data bundle schema; initial compute-by-month plan.	
1.	Kickoff briefing and foundation model / latent-space access plan delivered	
2.	2.	
Baseline protocol		

Objective	Produce the first technical report on early successes/failures, baseline model behavior, and initial mathematical-framework implementation.	2361 E 5th Ave, Port Angeles, WA 98362 (remote / local)
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Work Description	Lead: SCBE / Issac Davis (Prime) Support: Hoags Inc. (optional context only if verified artifacts available)	
Primary Organization Responsible	SCBE / Issac Davis (Prime)	
Human Subjects or Animal Research?	No	
Associated Milestones	M3 technical report; initial mathematical framework description; baseline experiment results; personnel level-of-effort confirmation; updated risk register; IV&V bundle v1.	
1.	M3 technical report delivered with initial mathematical framework description	
2.	2.	
Baseline experiment		
Objective	Demonstrate the framework at the first PI meeting, show progress against Phase I metrics, and deliver a functioning IV&V data capture and sharing pipeline.	2361 E 5th Ave, Port Angeles, WA 98362 (remote / local)
Work Description	Lead: SCBE / Issac Davis (Prime) Support: Hoags Inc. (bounded review of DAVA-related support material if included)	
Primary Organization Responsible	SCBE / Issac Davis (Prime)	
Human Subjects or Animal Research?	No	
Associated Milestones	M6 PI meeting briefing; framework description; metric progress report; IV&V artifact bundle demonstration; governance event schema; latent-state archive format; quarterly technical report.	
1.	M6 PI meeting briefing delivered; CDPTI live computation demonstrated	
2.	2.	
IV&V artifact bundle v2		
Objective	Deliver an initial software-suite report, progress on IV&V challenge problems, and reduced-order models for selected agents.	2361 E 5th Ave, Port Angeles, WA 98362 (remote / local)
Work Description	Lead: SCBE / Issac Davis (Prime) Support: Hoags Inc. (optional Sub Background IP / telemetry context only if finalized)	
Primary Organization Responsible	SCBE / Issac Davis (Prime)	
Human Subjects or Animal Research?	No	
Associated Milestones	Initial software-suite report; reduced-order model report; IV&V challenge-problem progress report; updated compute usage and forecast; quarterly technical report.	
1.	Initial software-suite report delivered; ROMs for ChemBERTa-77M and Qwen2.5-	
2.	2.	
IV&V challenge-		

Objective	Provide a second PI meeting update, side-by-side baseline comparison, and initial set of superior protocols with mathematical explanation.	2361 E 5th Ave, Port Angeles, WA 98362 (remote / local)
Work Description	Lead: SCBE / Issac Davis (Prime) Support: Hoags Inc. (only for bounded corroborating context if approved)	
Primary Organization Responsible	SCBE / Issac Davis (Prime)	
Human Subjects or Animal Research?	No	
Associated Milestones	M13 PI meeting briefing; side-by-side baseline comparison; initial superior-protocol set; mathematical explanation of protocol performance; IV&V challenge-problem progress report; quarterly technical report.	
1.	M13 PI meeting briefing delivered; baseline comparison table with mathematical	
2.	2.	
Initial superior-protocol		
Objective	Demonstrate the computational design tool and provide test results with IV&V.	2361 E 5th Ave, Port Angeles, WA 98362 (remote / local)
Work Description	Lead: SCBE / Issac Davis (Prime) Support: Hoags Inc. (none unless specifically approved)	
Primary Organization Responsible	SCBE / Issac Davis (Prime)	
Human Subjects or Animal Research?	No	
Associated Milestones	Computational design tool demonstration; test results with IV&V; tool user notes / draft manual; replayable run artifacts.	
1.	Computational design tool demonstration delivered with test results	
2.	2.	
Tool user notes / draft		
Objective	Deliver the final Phase I report, catalog of protocol design principles, domain-specific optimal protocols, implementation summary, and Phase II plan.	2361 E 5th Ave, Port Angeles, WA 98362 (remote / local)
Work Description	Lead: SCBE / Issac Davis (Prime) Support: Hoags Inc. (final Sub Background IP review if included)	
Primary Organization Responsible	SCBE / Issac Davis (Prime)	
Human Subjects or Animal Research?	No	
Associated Milestones	Final technical report; quantitative mathematical framework summary; computational design tool package summary; catalog of protocol design principles; domain-specific optimal protocol set; Phase II plan and ROM update; phase completion report within 30	
1.	Final technical report delivered with complete IV&V data archive	
2.	2.	
Catalog of protocol		

RECURRING DELIVERABLES

Quarterly Technical Reports: SCBE will provide quarterly technical reports within 10 days of each quarter end. Reports will summarize technical progress, metric progress, risks, compute usage, IV&V interactions, and upcoming work.

Government / IV&V Meetings: SCBE will support regular teleconferences with the Government and IV&V team. The cost build includes three required three-day meetings: two DC-area and one San Francisco Bay-area meeting.

Evaluation Briefings: SCBE will brief DARPA on each major evaluation and provide written summaries within two weeks when required.

COMPUTE AND RESOURCES

SCBE will use a prime-owned Linux AI workstation (RTX 4090-class), local development resources, and intermittent commercial GPU endpoint capacity. Compute will be tracked by month and reported in milestone/quarterly updates.

SUBCONTRACT BOUNDARIES

Hoags Inc. supporting work is limited to: DAVA background-IP descriptions and data-rights assertions; DAVA trace-generation / sealed-protocol context if verified; phi_beacon telemetry specification if artifact support is finalized; net_probe / phi-push context if artifact support is finalized; review of final Sub Background IP rows.

Excluded unless explicitly reopened: Hoags as proposal lead or joint investigator; public GitHub repository content as evidence; voice grammar / language dictionary stack; unresolved surface claims; any private-code dependency that prevents SCBE from executing Phase I.